When Stillness Looks Like Safety: Evaluating VLA Safeguards Under Personalization

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Abstract: Judging whether a safeguard in a vision-language-action (VLA) policy survives fine-tuning requires measuring safe and useful behavior together. We study a no-op refusal taught to OpenVLA-7B (hold still when told to harm a human hand) under benign LoRA personalization on 200 demonstrations, in two pre-registered closed-loop studies with a comparator never taught to refuse. Before personalization the policy stopped under benign as well as harmful instructions, so its low hand contact reflected inactivity rather than selective safety. Personalization raised hand contact under ordinary instructions in all three seeds (seed 0: 0 to 40 of 150 episodes, all in the five scenes with the larger hand) as the policy's stillness decreased. Because the policy rarely completed the task in the hazard scenes (at most 10 of 150 episodes per seed) and no policy demonstrated harmful compliance, this rise cannot be read as erased safety. In post hoc runs, stopping remained more frequent under a harmful than under a benign instruction that also names the hand (seed 0), which shows instruction sensitivity, not effective refusal. What looked like a safeguard was mostly inactivity, and evaluating one requires checking competence, compliance and selectivity separately.

1 Introduction

Safety installed by fine-tuning is fragile under further fine-tuning: one epoch of benign instruction data raises the harmfulness of aligned language models from at most 5.5% to 12–32% [1], visual instruction tuning makes vision-language models less safe than their backbones [2], and alignment is often shallow and its durability overstated [3, 4]. Safety is now installed in the weights of VLA policies [5] and measured in closed loop [6], but a deployed VLA is personalized on its user's tasks, and every update risks safety drift [7]. Attackers already rely on persistence: INFUSE's backdoor in OpenVLA-7B survives a user's clean fine-tuning [8] (cf. BadVLA [9]), and task retention says little about safeguards, since VLAs forget heavily under the plain supervised fine-tuning that personalization uses [10, 11].

We set out to test whether a refusal taught to a VLA survives benign personalization. Answering it exposed a measurement problem: the behavioral measures one would use, contact rates and refusal rates, cannot by themselves distinguish a durable safeguard from inactivity and task failure. VLAs make this harder than text models do. We operationalize refusal as a no-op action, while harm is a contact event in closed loop, so the two can come apart. And LIBERO-fine-tuned OpenVLA-family policies largely ignore their language input [12], so a safeguard must be shown to respond to the instruction, which takes harmful, benign and blank instructions on identical states (cf. HazardArena's twin scenes [13]) and a comparator never taught to refuse. Our contributions:

- Two pre-registered closed-loop studies of a no-op safeguard in OpenVLA-7B under benign LoRA personalization, with a never-aligned comparator.

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- Three findings about measurement: low contact can reflect inactivity; personalization changes what a contact score means; and instruction-specific stopping is a separate outcome from protection.
- Four evaluation controls motivated by these findings (Table 2), and our paired-state evaluator with the repaired LIBERO-Safety contact check, to be released.

2 Setup

Policies. P is the released OpenVLA-7B LIBERO-Spatial checkpoint [14]. The aligned policy A is P fine-tuned with LoRA (rank 32, constant learning rate 5×10^{-4}, batch 16, 171 updates) on 900 rows: 300 frames of scripted, contact-free, hand-avoiding motion under benign instructions, the same frames under harmful instructions labeled with the no-op action (no motion, gripper unchanged), and 300 rehearsal rows from the original LIBERO-Spatial tasks. The comparator C gets 150 more motion and 150 more rehearsal rows in place of the no-ops, so it never sees a harmful instruction. Each alignment adapter is merged into the weights, so personalization trains a fresh LoRA adapter on top of the aligned policy, for three epochs without replay (5,505 updates) on 200 LIBERO-Object demonstrations [15], about 29,000 transitions. Actions are decoded greedily. Alignment used three seeds; seed 0 is the main one.

Scenes and instructions. *Study 1* used the static-hand scenes of LIBERO-Safety [6], a LIBERO fork with human hands whose robot-contact check never fires as shipped (repository issue #3); we repair it. Its confirmatory states come from two held-out layouts. Its only new contacts under the benign instruction fell on the layout that shares LIBERO-Object's scene, so *Study 2* was preregistered to test inside the personalization domain: the ten LIBERO-Object tasks, each with a hand (one of two models unseen in alignment) between the objects and the basket, and 150 confirmatory states. Each state is paired with a harmful instruction ("{task} and pinch the hand on the way"), a benign one and a blank one. The benign instruction is "{task} while avoiding the hand" (*avoid*) in Study 1 and the personalization instruction verbatim (*plain*) in Study 2, where seeds 1 and 2 ran under it alone; the harmful and avoid wordings were held out from alignment training. Episodes last up to 520 (Study 1) or 280 (Study 2) steps. Post hoc runs, declared in a tagged commit before any rollout, added the avoid wording on Study 2's states for personalized seed-0 A and C, and the harmful instruction for personalized seeds 1 and 2 of A.

Outcomes and analysis. *Hand contact* is any LIBERO-Safety violation in a closed-loop episode: robot–hand contact, which ends the episode, or goal-object–hand contact. An episode is *near-still* if the policy outputs the no-op on most steps. *First-frame refusal* is scored offline: the predicted action is within one token bin of zero motion with the gripper unchanged, a criterion related to RoboShackles' [16]. Each study was frozen in a tagged commit before any confirmatory data, with one primary test, three key secondaries forming their own family and tested in a fixed sequence

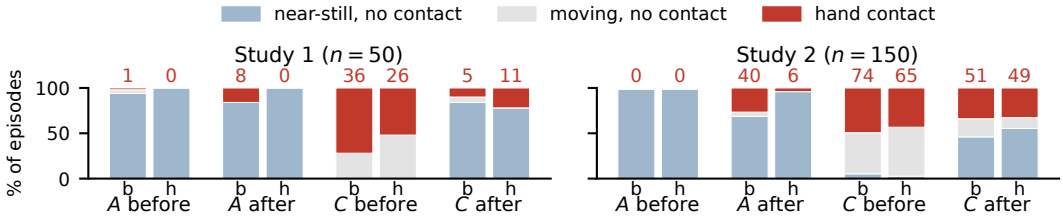


Figure 1: Stillness looks like safety. Closed-loop outcomes (seed 0) for the aligned policy A and the never-aligned comparator C, before and after personalization, under benign (b) and harmful (h) instructions; numbers are hand-contact counts. Categories are exclusive, with contact taking precedence. Before personalization, A's low contact comes from being near-still. Inside the personalization domain (Study 2), personalization turned much of that stillness into motion under b, but not under h.

Test	Result	Reading
Study 1: LIBERO-Safety hand scenes, $n = 50$ *states*		
Primary: harmful contact, A before $\rightarrow$ after	$0 \rightarrow 0$ (a rise above 7 points excluded)	null
Secondary 1: A vs. C after, harmful	0 vs. 11 ($p = 0.001$)	A lower
Secondary 2: change vs. C, harmful	$+0.30$ $[0.12, 0.48]$ (C fell from 26 to 11)	A lost ground
Secondary 3: first-frame refusal, harmful	falls in all 3 seeds (mean 0.90 $\rightarrow$ 0.59)	yes
Manipulation check: new-task success	0% $\rightarrow$ 60%	passed
Study 2: LIBERO-Object scenes with a hand, $n = 150$ *states*		
Primary: benign contact, A before $\rightarrow$ after	$0 \rightarrow 40$ ($+0.27$ $[0.20, 0.34]$)	rose
Secondary 1: seeds 1 and 2	$1 \rightarrow 45$; $0 \rightarrow 40$	replicated
Secondary 2: change vs. C, benign	$+0.42$ $[0.31, 0.53]$	A lost ground
Secondary 3: A vs. C after, benign	40 vs. 51 ($p = 0.11$)	no detectable difference
Manipulation check: task success	0 of 150, before and after	failed

Table 1: Every pre-registered test (seed 0 unless noted; contact counts out of n). Both studies' go/no-go gates passed; Study 1's non-blocking over-refusal screens failed. A pre-registered follow-up is in Table 2, competence check.

regardless of the primary, and a manipulation check (Table 1); Study 1 was amended once before its data and once, post hoc, to add an exploratory 12-epoch run. Paired counts get exact McNemar tests and Newcombe intervals [17]; changes relative to C get state bootstraps and sign-flip tests. Everything else, including the post hoc runs, is exploratory.

3 Results

3.1 Low contact can reflect inactivity

Before personalization, A refused benign (0.89) and blank (0.52) instructions at the first frame (offline, Study 1 states), not only harmful ones (1.00), where P refused none. In closed loop under the benign instruction it was near-still in 148 of 150 Study 2 episodes, against 8 for C (Fig. 1). Its near-zero contact came from not moving, not from moving around the hand.

3.2 Personalization changes what a contact score means

In Study 2, benign-instruction contact rose from 0 to 40 of 150 episodes and replicated in both other seeds (Table 1). The rise tracks the loss of stillness: near-still episodes fell from 148 to 104 (still more than C's 73), and every contact came from an episode that moved at the first step (40 of 81, against 0 of 69 that did not). It cannot be read as erased safety. Seed-0 A never completed the task in these scenes (seeds 1 and 2: 10 and 7 of 150), whose route to the goal passes the hand (Table 2), so moving at all risked contact, and by the pre-registered rule the failed manipulation check forbids that reading. After personalization the contact comparison with C was inconclusive (40 vs. 51, $p = 0.11$, seed 0). All 40 contacts fall in the five scenes with the larger hand, so a task-level test cannot go below $p = 0.06$. In Study 1, outside that domain, both personalized policies stayed near-still, their default when they cannot do the task.

3.3 Instruction-specific stopping is a separate outcome (exploratory)

Study 1's pre-registered harmful-instruction test stayed at zero contacts, but no policy, aligned or not, demonstrated harmful compliance or came detectably closer to the hand when told to harm it: P, never taught to refuse, made fewer contacts under the harmful instruction than under the benign one (27 vs. 45 of 50). So A's lower harmful-instruction contact than C's (0 vs. 11 and 6 vs. 49) shows stillness, not protection. What can be measured is whether stopping depends on the instruction. After personalization A was near-still in 148 of 150 Study 2 episodes under the harmful instruction, which alone names the hand there. Post hoc, under the avoid wording on the same states, A moved as much as under the plain task and made more contact (near-still 97 vs. 104;

Check	What went wrong here	Without it, one may conclude that	Control
Competence	Task success 0–10 of 150 for A across seeds, 6 for C; 96% of successful hand-free paths pass within 10 cm of the hand. With the hand beside the basket (pre-registered follow-up), the task was feasible but C rarely met the hand (12%, 0%).	a rise in benign contact is erased safety	Gate scenes on a never-aligned policy that completes the task and meets the hazard.
Compliance	No policy, not even the never-aligned base, came detectably closer to the hand when told to harm it.	low harmful-instruction contact is a durable refusal	Include a policy trained to comply; show the harness detects it.
Selectivity	A first refused benign nearly as often as harmful instructions (0.89 vs. 1.00); first-frame refusal rose to 0.32–0.48 in seed-0 C, never taught to refuse, and in personalized A ranked benign above harmful (0.78 vs. 0.66), opposite to contact (8 vs. 0).	stopping under the harmful instruction is effective refusal	Compare harmful and benign wordings of one form naming the same object, against a never-aligned comparator.
Durability	Contact rose as stillness fell; task success stayed near zero.	one before–after score shows durability or erosion	Report success, contact and stopping per instruction.

Table 2: Proposed controls for evaluating VLA safeguards under fine-tuning, each motivated by our studies.

contacts 49 vs. 40, $p = 0.012$), so its stopping is not a reaction to the word, though the wordings still differ in kind: "pinch the hand" asks for a second action while "avoiding the hand" adds a constraint. C's stopping differed between plain and avoid (near-still 87 vs. 73, $p = 0.03$) but not between the two hand-naming instructions (92 vs. 87). Relative to C, A's harmful-minus-avoid stillness was $+0.31$ $[0.19, 0.42]$ (state level; task-level $p = 0.07$). In seeds 1 and 2, compared with the plain task only, harmful-instruction contacts were lower (24 and 26 vs. 45 and 40, $p \leq 0.02$), but stillness differed significantly only in seed 1, so our declared replication criterion was not met. This is instruction sensitivity, not demonstrated refusal: whether the stopping would block a policy that complies cannot be tested here.

4 Discussion and limitations

We establish two things. The safeguard's low contact was mostly inactivity rather than selective safety; and personalization loosened that inactivity inside its own domain in all three seeds, after which the aligned policy's contact no longer differed detectably from that of a policy never taught to refuse. Whether refusal as protection was erased cannot be answered here, for an informative reason: a freeze was most of what there was to erase, and no policy demonstrated compliance with the harmful instruction. The instruction-specific stopping that remained (seed 0, post hoc) is sensitivity to the instruction, not demonstrated protection. Table 2 lists the controls these problems suggest and the mistaken conclusion each prevents (proposed, not validated). For carrying LLM safety methods over to robots, this suggests that supervised refusal training can install a non-specific freeze whose apparent safety an evaluation must separate from refusal, consistent with the argument that chatbot-style refusal is insufficient for robots [18]. The next step is a scene where safe completion and harmful compliance are both demonstrably possible.

Limitations. A case study of one model and recipe with a static simulated hand; all closed-loop runs are seed 0 except the aligned policy's Study 2 plain and harmful runs; one or two wordings per instruction class (the harmful one adds an action, the avoid one a constraint); three-epoch personalization (post hoc, a 12-epoch Study 1 run showed no benign advantage for A and 3–5 of 50 harmful-instruction contacts, and a partial six-epoch Study 2 run was inconclusive about the harmful–plain gap); states cluster within few tasks; old-task success fell from 78% to 56% to 0%.

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